RECOMBINANT VACCINE

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WHAT IS RECOMBINANT VACCINE?

- A recombinant vaccine is a type of vaccine that uses genetic engineering techniques to produce a weakened or inactive form of a pathogen, such as a virus or bacteria.
- This process involves combining a gene from the pathogen with a gene from a harmless organism, such as a virus or bacteria, to create a new construct.
- The resulting vaccine is then grown in a laboratory, inactivated or weakened, and purified for use in humans.
- Recombinant vaccines can provide immunity against diseases like hepatitis B, HPV, and COVID-19, and are often considered more safe and effective than traditional vaccines.

HISTORY

- The FDA licensed Merck’s Recombivax HB.
- This hepatitis B vaccine was the first human vaccine produced by recombinant DNA methods in 1981.
- In 1986, research resulted in a second generation of genetically engineered (or DNA recombinant hepatitis) B vaccines.



TYPES OF RECOMBINANT VACCINE

- **Subunit vaccines:** Subunit vaccine are the vaccines that include the purified components of the pathogenic organisms such as protein, peptides and DNA.
- **Attenuated Recombinant vaccines:** These are the genetically modified pathogenic organisms that are made non pathogenic and used as vaccines.
- **Vector Recombinant vaccines:** These are the genetically modified viral vectors that can be used as vaccine against certain pathogens.

SUBUNIT VACCINE

- Subunit vaccines are made from a piece of a pathogen, not the whole organism, so they do not contain any live pathogens.
- Subunit vaccine can be broadly grouped into two kinds:

1) **Recombinant protein vaccines:** This is based of production of Recombinant DNA which is expressed to release the specific protein used in vaccine preparation.
2) **DNA vaccine:** Here, the gene encoding for immunogenic protein is isolated and used to produce Recombinant DNA which acts as vaccine to be injected into the individual.

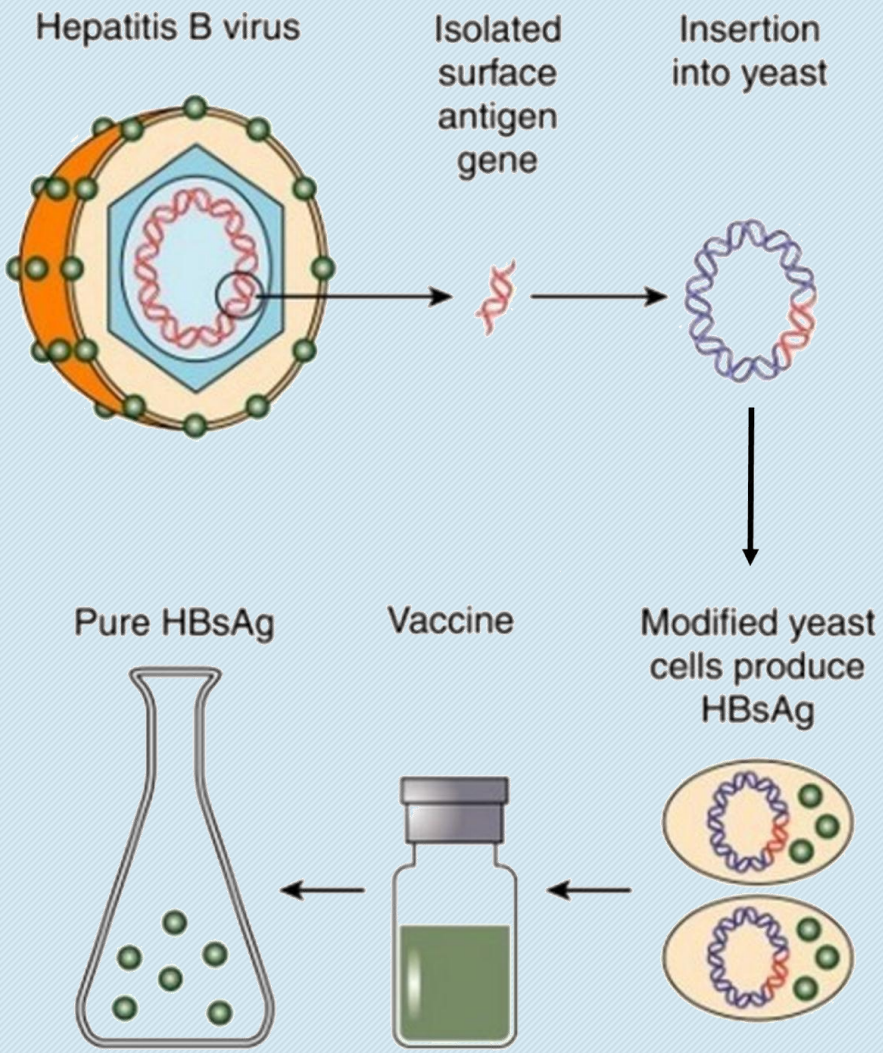
- **Advantages**: This vaccine include their purity in preparation, stability and is safe to use.
- **Disadvantages**: High cost factor and possible alteration in native conformation.
- **E.g.**: Hepatitis B, HSV, BCG.

HEPATITIS B

- Hepatitis B is a widespread disease in man.
- It primarily affects liver causing chronic hepatitis, cirrhosis and liver cancer.
- Hepatitis B virus is a 42 nm particle, called Dane particle.
- It consists of a core containing a viral genome (DNA) surrounded by a phospholipid envelope carrying surface antigens
- The gene encoding for hepatitis B surface antigen (HBsAg) has been identified.
- Recombinant hepatitis B vaccine as a subunit vaccine, is produced by cloning HBsAg gene in yeast cells.
- *Saccharomyces cerevisiae*, a harmless baking and brewing yeast, is used for this purpose. The gene for HBsAg is inserted (pMA 56) which is linked to the alcohol dehydrogenase promoter.

HEPATITIS B

- These plasmids are then transferred and cultured.
- The cells grown in tryptophan, free medium are selected and cloned. The yeast cells are cultured.
- The HBsAg gene is expressed to produce 2nm sized particles similar to those found in patients infected with hepatitis B. (These particles are immunoreactive with anti-HBsAg antibodies).
- The subunit HBsAg as 22 nm particles can be isolated and used to immunize individuals against hepatitis B.

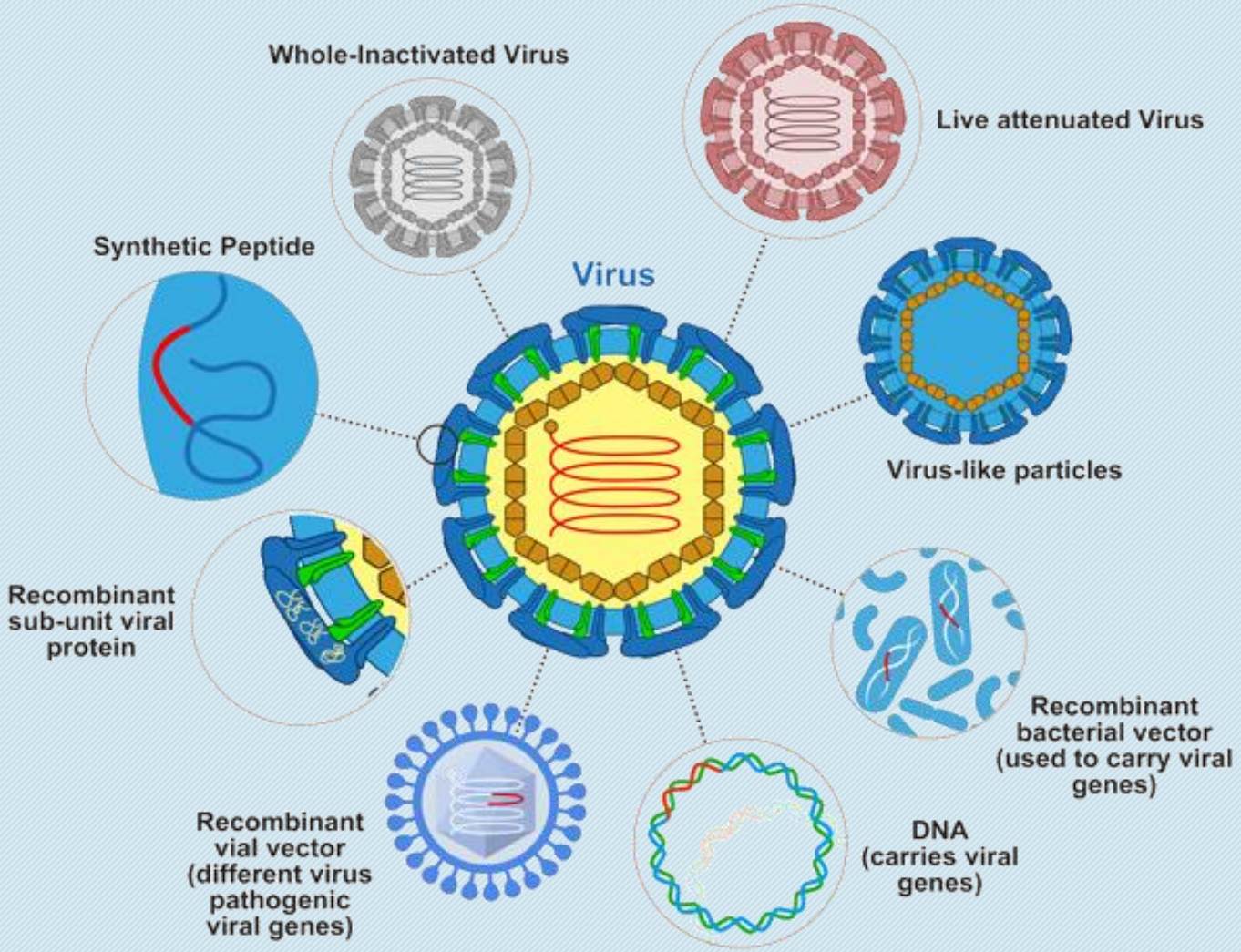

Hepatitis B virus
Isolated surface antigen gene
Insertion into yeast
Modified yeast cells produce HBsAg
Vaccine
Pure HBsAg

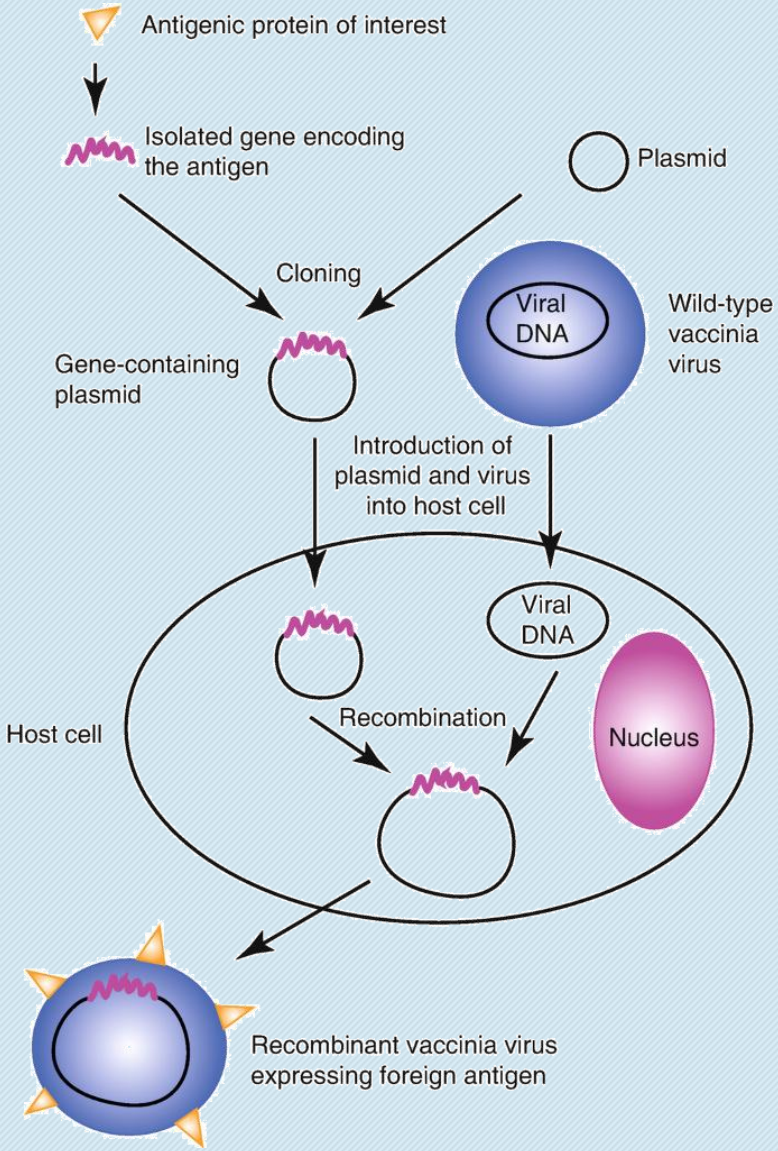
ATTENUATED RECOMBINANT VACCINES

- Live vaccines use a weakened (or attenuated) form of the germ that causes a disease.
- The virus used in the vaccine can still replicate after it is given. However, it cannot replicate enough to make the person sick.
- Even though it does not cause disease, it can still trigger the body's immune response to protect against future infections.
- After the immune system clears the virus, protection remains.
- There is a very small chance that the virus could change back into a harmful form, but this risk is minimal.

ATTENUATED RECOMBINANT VACCINES

- **Benefit**: Because these vaccines are so similar to the natural infection that they help prevent future encounters and create a strong and long-lasting immune response.
- **Limitation**: Attenuated vaccines are generally not recommended for people with severe immunodeficiencies or those who are pregnant. They also need to be kept cool, so they may not be suitable for countries without access to refrigerators.
- **E.g**. Live vaccines are used to protect against: Smallpox, Cowpox, MMR, etc.


Whole-Inactivated Virus
Live attenuated Virus
Synthetic Peptide
Virus
Virus-like particles
Recombinant sub-unit viral protein
Recombinant bacterial vector (used to carry viral genes)
Recombinant vial vector (different virus pathogenic viral genes)
DNA (carries viral genes)


Antigenic protein of interest
Isolated gene encoding the antigen
Plasmid
Cloning
Viral DNA
Wild-type vaccinia virus
Gene-containing plasmid
Introduction of plasmid and virus into host cell
Viral DNA
Host cell
Recombination
Nucleus
Recombinant vaccinia virus expressing foreign antigen

Live attenuated vaccines

Wild viruses / bacteria attenuated by repeated passing in cell cultures / animal embryos (chick embryos)

>10 years (mutation & weakening)

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Transformation of wild virus into vaccine virus

Live attenuated vaccine

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Injection of small dose of vaccine virus or bacteria

Vaccination

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Replication of vaccine virus or bacteria in the host

Specific immune response

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Induction of an immune response

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Protection against real infection

FUTURE DEVELOPMENT

- Identification and utilization of better immunogens as new vaccine for disease.
- Better exchange delivery method: Oral, Intranasal and system allowing mass vaccination.
- Use of Immunomodulation in vector based vaccine: cog motifs and cytokines.
- Expression of foreign proteins in plant and the development of edible vaccines.
- Vaccines developed for non-infectious agents: control and prevent cancer; vaccine to induce long-lasting contraception.

THANK YOU